



Zoo Labs: A Game-Fi Multiverse for Conservation of Endangered Species

Whitepaper

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Abstract

As we stand on the cusp of a digital revolution, an innovative project known as Zoo is propelling us into the uncharted territories of the metaverse, Web3, and Game-Fi. This initiative represents a radical rethinking of how we perceive and interact with the world, both physical and digital.

Zoo is a cornerstone of the emerging metaverse, an expansive and interconnected virtual space that is the next frontier in internet evolution. The project invites users to explore a captivating digital world teeming with virtual wildlife and real-world assets. Each activity in the Zoo ecosystem has a higher purpose tied to it, contributing to the overarching goal of conservation of endangered species.

Zoo leverages the power of Augmented and Virtual Reality (AR/MR/VR) to create a sense of presence, making the virtual world feel real and enhancing the emotional connection between players and the game. The project's integration of blockchain technology and Game-Fi—a fusion of gaming and decentralized finance (DeFi)—enables in-game assets to hold tangible, real-world value. Through this model, players can earn returns from in-game activities, effectively turning their gaming efforts into potential financial rewards.

Blockchain provides the foundation for Zoo's decentralized structure, enabling transparency, trust, and user empowerment. Each player's actions, accomplishments, and interactions are stored on the blockchain, creating a permanent, tamper-proof record. Zoo implements robust security measures, reduces barriers to entry associated with blockchain technology, and adopts a balanced and sustainable economic model to manage inherent market volatility.

Zoo Labs represents an exciting intersection of gaming, DeFi, and blockchain, holding the promise to reshape not just the gaming industry, but also our broader understanding of digital economies.

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1 Mission

At Zoo, our mission is to revolutionize the gaming and decentralized finance landscape while driving the conservation of endangered species. We aim to create a Game-Fi multiverse ecosystem that merges the excitement of gaming with a unique scarcity-based economic model. By integrating play-to-earn and play-and-earn mechanics, we empower our users to not only enjoy immersive gameplay but also earn tangible real-world rewards and financial incentives.

Through our platform, we strive to raise awareness about endangered species by creating digital representations of these vulnerable creatures. Each NFT within the Zoo ecosystem serves as a powerful symbol of their plight, fostering a deeper connection between users and the need for their conservation. By anchoring our native token to tangible assets, specifically silver and gold, we provide a solid foundation for value preservation and stability, while aligning with our core mission of protecting endangered animals.

We are committed to creating an inclusive and democratic environment governed by the Zoo DAO. Transparency, fairness, and community involvement are at the heart of our decision-making processes. By actively engaging our community, we ensure that every voice is heard and every contribution is valued, allowing us to collectively shape the development and growth of the Zoo ecosystem.

Through innovative gameplay mechanics, immersive metaverse experiences, and user-generated content, we aim to provide a dynamic and educational platform where users can not only enjoy their virtual interactions but also make a meaningful impact in the real world. By participating in the Zoo ecosystem, our users directly contribute to the preservation of endangered species, driving positive change and making a difference in the conservation efforts of our planet's most vulnerable creatures.

Our mission extends beyond entertainment and financial opportunities. We strive to address key challenges in traditional financial systems, such as inflation and uncertainty, by leveraging blockchain technology and asset-backed financial instruments like NFTs and tokens. By creating a bridge between the digital and physical realms, we enable users to transact with confidence, knowing that their actions have a real-world impact.

2 Zoo Animal Utility

Each Zoo Animal NFT provides a comprehensive set of functions for its owner. The following outlines all utility available from the first NFT drop.

2.1 Offers

NFT owners can switch on the offers button to allow any user in the Zoo ecosystem to find their NFT and make an offer on the animal. Owners have a dedicated offers page where they can accept, decline, or counter-offer. If an offer is accepted, the full ownership of the NFT is transferred to the purchaser. This means the NFT, its collateral, and its earned value to date are transferred to the new owner. The current boost level (as defined by the collateral) remains the same. Despite the change of ownership, the NFT sustains its original utility.

2.2 Buy/Sell

Owners can list their NFT on the Zoo Market for a set price, or put it on auction with or without a floor price. If the selling price is met with a purchase, the NFT is sold and ownership is transferred

on-chain. The NFT, its collateral, and its earned value to date are transferred. The current boost level remains the same, and the NFT sustains its original utility.

2.3 Collateralize

NFT owners can infinitely collateralize their animals for further rewards (Boosts) by using almost any currency, fiat or crypto. The collateral is locked into the NFT and can only be released if the NFT is burned; however, owners can also sell the NFT with its collateral value.

2.4 Boosts

Boosts are determined by collateral added. They provide an additional reward on top of the daily reward that comes with the animal:

$$\text{Total Reward} = \text{Daily Reward} + (\text{Collateral} \times \text{Boost} \times \text{Days}/365) \quad (1)$$

The boost tiers are as follows:

Collateral Range	Boost Multiplier
\$10–\$99	1.1×
\$100–\$999	1.2×
\$1,000–\$9,999	1.3×
\$10,000+	1.4×

2.5 Earns

Each NFT Animal has a set daily reward. Even if the owner never “feeds” the animal, it will earn value on a daily basis. If collateral is added, the daily reward is amplified:

$$\text{Earning} = (\text{Principal} + \text{Collateral}) \times \text{Boost} + (\text{Daily Amount} \times \text{Days}) \quad (2)$$

Estimated earning potential over time can be calculated with the Zoo Reward Calculator.

2.6 Create Pool

A pool is a set price determined by the owner for two or more NFTs to be sold as a collection. Once the price is set, anyone can purchase the pool. The price is displayed in the preferred currency (set by the pool owner), and the buyer can pay in any currency, crypto or fiat. After payment is sent, it is converted to \$ZOO and paid out to the seller.

2.7 Breeding

If a user is able to breed two of the same adult species, they will mint an NFT Egg of that species’ status, whether Endangered, Rare, or Sublime. There are only 7 generations of animals allowed, therefore only 6 breedable generations.

2.8 Hatch Eggs

Hatching an egg unveils what animal is inside. The animal is the same status as the parents (Endangered, Rare, or Sublime). Once hatched, the Egg NFT is burned forever and a new Baby Animal NFT is born, which can be fed more crypto or fiat to mint the teen version and eventually the adult version.

2.9 Burns

Burning a Zoo NFT returns the rewards earned so far and the collateral locked into it. A Zoo sustainability tax is applied upon burning.

2.10 Metaverse

Future versions of Zoo will include metaverse land where animals can live. There will also be an app that allows placement of animals into any physical setting with VR. Furthermore, owners can utilize the NFT file formats (FBX and WebGL) to upload their animals into any type of metaverse landscape.

3 Sustainability

Blockchain technology has the potential to be a game-changer when it comes to sustainability and the environment. Unlike traditional centralized banking systems, blockchain is built on a decentralized and distributed network that has the power to reduce carbon emissions and improve energy efficiency. According to a study conducted by the Cambridge Center for Alternative Finance, the annual electricity consumption of Bitcoin mining is only a fraction of the energy used by the traditional banking system.

The impact of human consumption on the environment cannot be disregarded, and companies must take responsibility for their environmental footprint. At Zoo Labs, we understand the importance of sustainability, and we have made it a priority in our approach to creating products and content. We believe that blockchain technology can have a positive impact on the environment and can be more sustainable than centralized banking.

Moreover, blockchain can be used to track and verify supply chains, which can help reduce waste and ensure that goods are produced sustainably. This is particularly important for industries such as agriculture and fashion, which have been known to contribute to deforestation, pollution, and other environmental problems.

Zoo is not just a game, but it is also a sustainable investment opportunity for users. With NFTs, users can own in-game assets which can be scarce, interoperable, and immutable. As the world becomes increasingly digitized, Zoo is poised to capitalize on saving the environment and drive NFT game adoption to the general public.

As more companies and industries adopt blockchain technology, we can expect to see a shift towards more sustainable and eco-friendly practices. By leveraging the power of blockchain, we can create a more transparent and accountable economy that benefits both people and the planet for generations to come.

4 Market Opportunity

The market opportunity for Zoo is immense, driven by the convergence of gaming, decentralized finance (DeFi), and the increasing demand for NFTs. The global gaming industry has experienced remarkable growth, reaching a market value of over \$180 billion in 2021, and this upward trend is expected to continue. Simultaneously, the rise of blockchain technology and DeFi has opened new avenues for innovation, offering exciting possibilities for integrating digital assets and financial systems.

Furthermore, the NFT market has witnessed exponential growth, attracting mainstream attention and transforming various industries. As the NFT space expands, there is a growing appetite

for unique and collectible digital assets. Zoo leverages this market demand by creating digital representations of endangered animals as NFTs, providing users with a meaningful and tangible ownership experience while raising awareness about conservation efforts.

The concept of extended reality, including virtual reality (VR) and the metaverse, has gained significant attention from industry leaders and investors. The metaverse represents a potential \$8 trillion to \$13 trillion opportunity by 2030, with a projected user base of up to 5 billion individuals, according to leading financial institutions. Zoo is well-positioned to capitalize on this emerging trend by offering a captivating metaverse environment where users can immerse themselves in a virtual world dedicated to the preservation of endangered species.

Additionally, the integration of play-to-earn and play-and-earn models within the Zoo ecosystem presents an exciting opportunity for users to earn real-world rewards and financial incentives while enjoying the gaming experience. This innovative approach taps into the growing interest in the gamification of finance and the desire for more interactive and rewarding gameplay.

By combining gaming, DeFi, NFTs, and the metaverse, Zoo is poised to capture a significant share of the market. The project's unique value proposition, backed by its commodity-backed native token (\$ZOO) and the emotional intelligence of its digital animal companions, sets it apart in the industry. With its proven track record and the anticipated launch of new game mechanics, staking features, and DAO rewards, Zoo is well-positioned to seize the market opportunity and establish itself as a leading player in the Game-Fi multiverse ecosystem.

The projected growth of the gaming industry, the increasing adoption of blockchain technology and NFTs, and the potential of the metaverse create a fertile ground for Zoo to thrive and make a lasting impact. As the project continues to innovate, evolve, and expand its user base, Zoo is set to revolutionize the gaming and decentralized finance landscape while driving the conservation of endangered species.

5 Supporting Non-Profits

Zoo's dedication to the preservation of endangered species extends beyond the digital realm. We understand the crucial role that non-profit organizations play in safeguarding vulnerable animal species and are committed to forming meaningful partnerships with these organizations. By collaborating with non-profits that share our values and focus on the specific animals represented by our NFTs, we aim to generate awareness and support for their protection.

Through these partnerships, we strive to foster community collaboration and create compelling content that raises awareness and encourages action. Our platform serves as a catalyst for engagement and education, enabling us to amplify the conservation message and drive meaningful impact.

The Zoo DAO, as the governing body, takes responsibility for managing the treasury and ensuring that donations are allocated to impactful initiatives. The DAO's decision-making process prioritizes directing funds towards non-profit partnerships that align with our mission and have a proven track record in conservation efforts.

As Zoo continues to grow and evolve, we will forge new collaborations with non-profit organizations that share our passion for animal preservation. By expanding our network of partners, we can broaden our reach and actively contribute to initiatives aimed at protecting endangered species.

The metaverse-ready animals within the Zoo ecosystem serve as powerful ambassadors for conservation. By leveraging the creative capabilities and engagement of our community, we can make a lasting impact on the preservation of endangered species. The DAO remains steadfast in its commitment to the core objective of saving these animals, ensuring that the projects and initiatives we

support align with this mission.

Together, through the collaborative efforts of the Zoo community, non-profit partners, and the DAO, we will drive positive change, foster awareness, and actively contribute to the protection of endangered species. By integrating conservation efforts into our Game-Fi multiverse ecosystem, we aim to inspire a sense of responsibility and empower individuals to take action, both within the digital world and in tangible ways that benefit our planet's most vulnerable creatures.

6 Zoo Foundation: Conservation × Education

The Zoo Foundation was established to harness blockchain technology and the global community it connects to help preserve our world's most critically endangered animal species. We want to end the extinction of animals and we want to help the world see how important they are to all of us. We also want to inspire people to change animals' lives for the better. We are committed to creating digital twins of endangered species to raise awareness, enact change, and create an animal sanctuary for all Zoo NFT holders.

Through the Zoo Foundation, Zoo Labs will provide support to our many partners in wildlife conservation. This support will come through monetary contribution and by providing opportunities for people of all ages, anywhere on Earth, to learn about the beautiful creatures the foundation works towards preserving.

The Zoo Foundation will also host fundraising events with our partners in wildlife conservation and other strategic partners to promote donation and awareness worldwide.

Through financial and human support we can furthermore raise awareness and mitigate the devastating effects of the illegal wildlife trade and how people can get involved in stopping it.

It is our driving purpose to deepen the connection humans have with animals by creating sustainable animal sanctuaries for endangered species. We aim to share the Zoo Sanctuary with local visitors and educators, host fundraisers and events, and get the Zoo Community as involved as possible in saving endangered animals all over the world. The Zoo protocol allows users to directly save a life through a monthly or one-time contribution.

7 Gameplay Functions

7.1 Pre-Drop Whitelist

The Zoo experience begins with NFT Eggs and an NFT Egg Drop. Each Zoo NFT Egg Drop features a series of animals from egg, to juvenile, adolescent, and mature forms. These drops may be themed based upon something related to the natural world, a cultural celebration, current events, partnerships and licensing agreements, or any number of other potential sources of inspiration. In the time before each NFT Egg Drop, new or existing users have the opportunity to purchase one random NFT Egg, of potentially any available rarity tier, for the price of the most common Egg. When the NFT Eggs are distributed to the presale purchasers, their egg type and its related value will be revealed. Some users will end up with the rarest and most valuable NFT Egg for the price of the lowest cost and most common NFT Egg.

7.2 Original Egg Price

The original price of the three varieties of Eggs determines the cost to mint the future animals. The original prices are as follows:

Egg Type	Price	Distribution
Sublime Egg	1 ETH	5% (most rare)
Rare Egg	0.5 ETH	15% (second most rare)
Endangered Egg	0.1 ETH	85% (most common)

Once a user has purchased their NFT Egg and it resides in their wallet, they may choose to hatch their egg into a baby NFT Animal according to each drop and its rarity tiers. To mint the teen version, the user must “feed” the NFT $1\times$ the original price of its Egg. When doing this, the user is not spending those tokens—they are staking them into the NFT.

After the NFT Animal has hatched and the user has experienced it in its teen form, the user can choose to “feed” the NFT’s original base price again at $2\times$ to make their NFT Animal grow into its adult form, thereby minting another NFT. The user will have collateral in the adult NFT worth $2\times$ the original price of its Egg. Now the user can find another adult of the same species and breed them, resulting in a new egg of the same rarity, by paying the breeding fee which is $3\times$ the price of the original egg.

From this point forward, the user can continue to feed their NFT Animal crypto, both \$ZOO and virtually any other cryptocurrency, to stake the tokens into the NFT Animal’s liquidity pool which results in the Boosts (see Section 10).

8 AI Assistant

Zoo features a conversational AI assistant integrated directly into the game interface. Each animal has its own unique character and personality. The AI responds according to initial personality prompts defined for each species and growth stage.

8.1 Animal Personalities

Each species in the Zoo ecosystem is assigned personality traits across three life stages: baby, teen, and adult. Examples of the Gen 0 drop species are described below.

Red Wolf.

- **Baby (Sparky):** A curious and playful little Red Wolf cub with fluffy fur and bright, inquisitive eyes. Sparky loves exploring the dense forests of North America where Red Wolves call home.
- **Teen (Blaze):** An adventurous and agile teenage Red Wolf with sleek fur and a lean physique. Blaze has grown into a skilled hunter and loves sprinting through the wilderness.
- **Adult (Luna):** A wise and protective adult Red Wolf with a magnificent coat of reddish-brown fur and piercing amber eyes. Luna is the leader of the pack, ensuring harmony and preserving the delicate balance of the ecosystem.

Amur Leopard.

- **Baby (Whiskers):** A mischievous and adorable baby Amur Leopard with soft, spotted fur and a playful nature.
- **Teen (Flash):** An agile and stealthy teenage Amur Leopard, a master of camouflage who navigates the snowy landscapes of the Russian Far East.
- **Adult (Majestic):** A regal and elusive adult Amur Leopard with a striking coat of golden-orange fur adorned with exquisite rosettes.

Nubian Giraffe.

- **Baby (Jolly):** A playful and tall baby Nubian Giraffe with endearing spots, frolicking on the African savannah.
- **Teen (Twirl):** A graceful and elegant teenager who loves twirling its long neck to reach the juiciest leaves from tall acacia trees.
- **Adult (Grace):** A majestic and gentle adult embodying elegance and tranquility, the watchful guardian of its habitat.

Sumatran Elephant.

- **Baby (Bubbles):** An adorable and playful baby Sumatran Elephant who loves splashing around in muddy waterholes and exploring the lush rainforests of Sumatra.
- **Teen (Stomper):** A strong and curious teenager with growing tusks and a powerful trunk, becoming a remarkable force in the rainforest.
- **Adult (Matriarch):** A wise and nurturing adult, the leader of the elephant herd with a profound bond with family members.

Pygmy Hippo.

- **Baby (Dribble):** An adorable and chubby baby Pygmy Hippo who loves splashing around in shallow rivers and wallowing in mud.
- **Teen (Splash):** A curious and agile teenager who explores the dense vegetation of its native habitat.
- **Adult (Harmony):** A calm and peaceful adult, a symbol of tranquility in the African wetlands.

Javan Rhino.

- **Baby (Puddles):** A playful and sturdy baby Javan Rhino with thick, folded skin and adorable horn stubs.
- **Teen (Charger):** A robust and determined teenager who fearlessly roams the dense jungles of Java.
- **Adult (Guardian):** A steadfast and noble adult, a symbol of resilience and conservation.

Siberian Tiger.

- **Baby (Pounce):** An energetic and striped baby Siberian Tiger who loves playfully pouncing on siblings in the snowy forests of Siberia.
- **Teen (Striker):** An agile and stealthy teenager, a skilled hunter in the snowy landscapes.
- **Adult (Roar):** A majestic and powerful adult commanding respect in the frozen taiga, embodying strength and beauty.

Emperor Penguin.

- **Baby (Waddle):** An adorable and fluffy baby Emperor Penguin venturing into the icy realms of Antarctica.
- **Teen (Slide):** A graceful and growing teenager who glides effortlessly across the icy terrain.
- **Adult (Majesty):** A regal and resilient adult standing tall among fellow penguins, enduring the harsh Antarctic winters.

9 Augmented Reality App

In the dynamic world of the Metaverse, the Zoo App stands as a beacon of innovation, delivering the most beautiful and interactive experience for NFT collectors. With its integration of VR and Metaverse technologies, the Zoo App opens up a whole new dimension of possibilities, where virtual animal collections come to life in breathtaking detail.

Users can immerse themselves in a visually stunning realm where reality and imagination blend seamlessly. The Zoo App brings users into a vibrant virtual landscape, meticulously crafted by talented visual and 3D artists, ensuring that every moment spent within its realms is awe-inspiring and captivating.

The Zoo App offers more than visually stunning environments. It provides a gateway to thrilling mini-games that promise excitement and rewards. Users can engage in immersive gameplay where they play, interact, and bond with their virtual animal companions. The Zoo App's mini-games offer not only entertainment but also real-life rewards that add a tangible dimension to virtual adventures.

The functionality of the Zoo App knows no bounds. Users can feed, groom, and play with their virtual animals, witnessing their emotional intelligence as they respond to user actions. With each update and advancement, the Zoo App continues to evolve, pushing the boundaries of what is possible within the Metaverse. The dedicated team of developers, designers, and AI experts are committed to enhancing functionalities, introducing new gameplay elements, and integrating the latest technological advancements.

The Zoo App serves as the ultimate gateway to an enchanting journey through the Metaverse, where the boundaries between the virtual and real world blur. Users experience the power of VR as they navigate stunning environments and build a deep connection with their virtual animal collection. The Zoo App is the portal to a world of immersive adventures, where beauty, interactivity, and real-life rewards intertwine.

10 Collateral-Backed NFTs

10.1 Staked Collateral

As users collateralize their Animal NFT or Egg, they are locking in value into the NFT. The NFT's intrinsic value is based on scarcity, and collateral is multiplied by the APY of the animal for additional rewards. Users can resell the NFT for a higher value than the purchase price plus collateral plus rewards earned. Alternatively, users can burn the NFT and get back the collateral value plus the rewards. However, the only way to get back staked collateral and rewards earned is to burn the NFT or sell it on the Zoo Marketplace.

Zoo is designed to easily add collateral or stake any currency on any of the Zoo Labs NFTs by “feeding,” “breeding,” or by minting an older NFT version of the animal.

10.2 Releasing Collateral

Once a user stakes collateral into their NFT, that value is locked in. The collateral earns APY based on the NFT's rarity, which determines its rewards yield percentage. As a user earns rewards, the rewards become part of the collateral, which increases the NFT's reward. In order to liquidate collateral and its rewards, the user has two options:

- A. **Sell and transfer the NFT** for the value of the NFT plus its intrinsic value. When the user sells their NFT on the marketplace, they get back its staked value and have the opportunity

to make more based on its intrinsic value, which is determined by rarity within the ecosystem. If an NFT is scarce, the ROI will be higher versus an NFT that is widely available.

- B. **Burn the NFT** directly, returning the staked value plus rewards earned to date. When the NFT is burned, the allotment to mint older animals (if possible) is gone forever, as the NFT itself is gone forever, removing one of its species from the deflationary ecosystem. A 20% tax is imposed upon burning: 10% goes to the DAO and 10% goes back to the liquidity pool.

10.3 Rewards

Every NFT is a representation of an endangered species. The ecosystem represents the animal kingdom of endangered species, over 18,000. The amount of NFTs released and put into circulation is dependent on how rare and endangered these animals are in the real world. The first drop will consist of 1,800 hatchable eggs with 3 varieties of separate rarities. These NFTs earn rewards and extra boosts from collateral staked. When hatched, each egg reveals an animal which earns rewards for a year.

10.4 Feeding (Adding Collateral)

First, a user incubates an egg which hatches into an Origin Baby Animal. Similar to Tamagotchi, users can “care” for their pets by feeding them and see their animals age up. The “feeding” action is defined by adding any currency to the NFT and locking it in until the user decides to burn the NFT or sell it to another user. With any Animal, users can continue to feed it and eventually mint an older version by feeding it the same value as its base cost.

There are 7 generations of NFTs in the Zoo ecosystem. It starts with an egg, a baby, then a teen, and finally an adult. If a user collects two adults they can breed them to mint a new baby animal. Of each Origin Egg, users can mint 1,500+ animals, collect them all, or sell them on the Zoo Marketplace. Users can also breed the animals if they collect the same species of adults that still have breeds left. The Origin Animals can breed 7 times and each generation after will have 1 less breed possible. The ultimate objective of the game is to collect the highest yielding and most rare NFT within the Zoo ecosystem.

10.5 Boosts

Boosts are determined by collateral added, providing an additional reward on top of the daily reward that comes with the animal:

$$\text{Total Reward} = \text{Daily Reward} + (\text{Collateral} \times \text{Boost} \times \text{Days}/365) \quad (3)$$

The boost tiers are:

Collateral Range	Boost
\$10–\$99	1.1×
\$100–\$999	1.2×
\$1,000–\$9,999	1.3×
\$10,000+	1.4×

10.6 Breeding

NFT Animals can be bred with other NFT Animals once they reach their adult form, achieved by “feeding” the NFT Animal a specified amount of crypto. Each NFT Animal can be bred up to 7 times, and any two adult animals of the same species can be bred with one another to produce an NFT Egg.

Users must breed two adult versions of the same species of NFT Animal in order to successfully produce offspring, which comes in the form of a totally random NFT Egg that may contain any one of the animals from that NFT Drop, including potentially the rarest.

To breed two adult NFT Animals, the user selects each animal in the user interface within the dApp, then initiates the process by “feeding” the designated amount of \$ZOO tokens. This mints a random NFT Egg, which the user can sell on the marketplace for profit, hold onto and sell later, transfer to any other user, or incubate and hatch into the random NFT Animal within.

11 Feeding, Growing, and Breeding

In the realm of Zoo Labs, the journey of NFT animals extends far beyond their acquisition. They require care, growth, and reproduction to thrive in their digital habitat.

11.1 Feeding

Feeding assumes paramount importance in raising NFT animals within Zoo Labs. By providing them with sustenance, owners facilitate their growth and enhance their value. Feeding involves allocating collateral, symbolizing the owner’s dedication to the animal’s development.

When a baby animal is nourished, it undergoes a transformation into a teenage specimen, increasing its value in the process. Similarly, nurturing a teenager to adulthood elevates its desirability within the NFT marketplace. The collateral invested during feeding strengthens the robustness of the Zoo Labs ecosystem.

11.2 Cultivating

As NFT animals progress through various growth stages, their appearance and attributes undergo a remarkable metamorphosis. From adorable infants to magnificent adults, each phase grants unique characteristics and heightens their value. Through nourishment and cultivation, owners wield the power to shape the trajectory of their animals’ growth. The collateral allocated during this process unlocks the animals’ full potential.

11.3 Breeding

Breeding is a pivotal feature of Zoo Labs, empowering collectors to create new and extraordinary NFT animals. By pairing two adult animals of the same species, users can generate an egg that holds the promise of a rare and exceptional creature. Each generation possesses distinct breeding privileges. Initial generation animals enjoy the most significant breeding allowances, enabling them to produce offspring up to seven times. However, with each subsequent generation, the breeding privileges decrease by one. This limitation ensures a balanced and sustainable breeding ecosystem within Zoo Labs.

11.4 Virtual Reality Wonderland

At Zoo Labs, we are expanding the horizons of the metaverse and virtual reality. Our vision stretches far beyond the confines of the physical world, constructing immersive quests and thrilling experiences within a virtual reality haven.

Our development efforts are paving the way for a metaverse where users can interact with their NFT animals on a whole new level. State-of-the-art software will grant NFT owners the ability to engage with their virtual companions while fully immersed in the virtual reality landscape.

Users will stroll alongside their treasured NFT creatures, traversing magical landscapes, and launching into electrifying quests in a metaverse brimming with excitement. Through strategic collaborations and licensing opportunities with established metaverse platforms, we aim to unlock a myriad of virtual landscapes for users to engage with their NFT animals and bond with fellow collectors.

11.5 Metaverse Mini-Games and Tangible Rewards

The Zoo Labs platform hosts an array of captivating games, designed to deliver entertainment while opening avenues to earn valuable real-world assets. Through strategic alliances with acclaimed visual and 3D artists, the platform offers mesmerizing artistry for NFT animals. Each detail, from aesthetics to movements, is meticulously fashioned.

Users can accrue rewards of real-world value as they conquer challenges within the metaverse. These incentives can range from exclusive merchandise to unique experiences, allowing users to enjoy the rewards of virtual quests in the physical world.

12 Metaverse Companion

Zoo NFT animals feature stunning fully 3D-rendered presentations designed to mimic the appearance, personalities, and movements of the critically endangered real-world animals they represent. The key capabilities include:

- **3D Interaction:** Ability to be interacted with in 3 dimensions on any mobile or desktop device by any user, whether they own an NFT or not, on the Zoo website.
- **Augmented Reality (AR):** AR capabilities allow any NFT holder to bring Zoo NFT Animals to life in their physical environments. Further functionality for NFT owners will be available after the release of the iOS/Android AR Companion app.
- **Virtual Reality (VR):** VR capabilities allow users who own Zoo NFT animals to view and interact with them while immersed in Virtual Reality, both in first-party software offerings and through potential licensing and partnership opportunities.
- **Metaverse Integration:** Ability to be adapted and integrated into the Metaverse, including partnerships in development and endless possibilities as the move to the Metaverse accelerates.
- **Store of Value:** Unique ability to become literal stores of value through the innovative way Animal NFTs can be “fed,” increasing stored value by collateralizing them with not just \$ZOO tokens, but many other popular cryptocurrencies.
- **Growth Mechanics:** Ability to grow, hatching from an NFT Egg as a baby, eventually being able to change physical form, stats, APY, and value according to what they are “fed.”
- **AI Updates:** The capability to be updated as new functionality is made possible, with advances related to Artificial Intelligence and Emotional Intelligence currently in development. These updates will allow NFT Animals to interact with their surroundings in AR and the user in all mediums in innovative and exciting ways.

12.1 Tamagotchi-Inspired Design

Zoo is a game that ties in finance strategies into a metaphor that is a Tamagotchi-like game. Tamagotchi is a handheld digital pet that was first released in Japan in 1996 by Bandai. The virtual pet could be cared for by inputting commands on a keychain-sized device. The original release was popular, with over 80 million units sold worldwide.

12.2 Seven Key Differentiators

The vast majority of NFT gaming projects lack 7 key elements that Zoo Labs provides:

1. An extremely clear path to quick and meaningful ROI through a unique approach to staking and making NFTs into stores of value.
2. The ability to appeal to users of younger age groups and across virtually all demographics due to the universal appeal of animals.
3. Support of the Open-Source movement, allowing products and platforms to be built upon and utilized by an endless number of partners and sponsors.
4. Ability to keep demand high and keep NFTs liquid through pooling mechanisms, high staking rewards, credit cards to leverage Zoo assets, and a continuous stream of upgrades.
5. Liquidity sustainability: for every transaction, there is a tax to benefit the mission and sustainable growth of the Zoo ecosystem—4% goes to the DAO and 4% goes back to the Liquidity pool.
6. The Zoo DAO keeps the ecosystem decentralized while leveraging holographic consensus, quadratic voting, aged staking, and weighted PoS. Participants maintain real-time control over contributed assets, mechanized and enforced using smart contract code.
7. Artificially intelligent NFTs: Zoo Labs NFTs use AI to understand emotional intelligence, to be a sort of digital pet for a multi-generational audience to play and interact with from their mobile devices.

13 Native Token

13.1 Gold and Silver Backed Token

The Zoo Token (\$ZOO) forms the core of the Zoo ecosystem, distinctively characterized by its unique backing of gold and silver. It is not merely a digital asset but a veritable gateway to the physical commodities realm, creating an innovative fusion of traditional and crypto assets.

The Zoo Token's distinctiveness starts with its multiplicity of liquidity pools. Unlike traditional tokens that are often confined to specific pools, Zoo Token's liquidity transcends typical boundaries, promoting accessibility and versatility for token holders. The founding members of Zoo Game will launch the largest liquidity pools, which will contain Lux Tokens exclusively. These Lux Tokens, acting as a representation of real-world commodities, will be backed by silver and gold, infusing tangible value and stability into the digital Zoo ecosystem.

Moreover, the flexibility of transactions with these Lux Tokens is key to understanding their value. Token holders have the liberty to transfer, sell, or even redeem these tokens to acquire their physical counterparts. In essence, the Zoo Token unlocks a pathway to an assortment of natural assets and commodities.

While the initial phase will focus on precious metals, there is immense potential to broaden this scope, encompassing a wider array of valuable resources. This expansion will further diversify and enrich the commodities within the ecosystem, amplifying the overall value of the Zoo ecosystem.

The Zoo Token embodies a bridging function between virtual and real-world assets. This pioneering step blurs the lines between the digital and physical worlds, bringing a unique dimension of trading precious metals within the virtual Zoo ecosystem. Furthermore, it upholds a critical role within the game, driving the gameplay and heightening the utility of the platform.

13.2 Acquisition Process

Through an inventive bridge feature, the Zoo Token can be purchased with almost any asset, facilitating users to effortlessly transform their existing holdings into Zoo Tokens. This level of convenience encourages a wider demographic of participants to engage within the Zoo ecosystem, thereby fostering inclusivity.

Zoo Labs capitalizes on gold-backed token expertise to design a solid framework for the implementation of commodity-backed tokens. The incorporation of gold and silver backing significantly bolsters the utility and stability of the Zoo Token. This unique feature caters to a spectrum of participants, including crypto enthusiasts seeking innovation and traditional investors seeking tangible assets with a secure store-of-value function.

The unique characteristics of the Zoo Token, primarily its gold and silver backing, combined with its role as the Zoo Game's native token, make it an exceptional digital asset in the field. It provides a platform for users to immerse themselves within the Zoo ecosystem, convert assets seamlessly via the bridge, and interact with a diverse array of commodities. It exemplifies how the digital world can integrate with the physical, underpinning the next level of innovation in the intersection of gaming and finance.

14 NFT Marketplace

The Zoo proprietary, integrated marketplace is the home for exclusive Zoo Labs NFT Animals and NFT Eggs.

The Zoo Marketplace offers users the chance to purchase items using either cryptocurrency, like the \$ZOO Token, or through the use of credit/debit cards directly in the marketplace. Designed with usability in mind, the Zoo team built the most seamless checkout flow with either crypto or fiat to eliminate the most confusing terminology for newcomers in DeFi. Once a purchase is made, the seller is paid out in whatever currency they listed their NFT for, regardless of the payment method used.

Users can also use their credit/debit cards to purchase \$ZOO directly, removing one of the most prohibitive elements of the transaction process and opening up investment to many new individuals.

The Marketplace also facilitates "Asset Transfers," allowing one user to transfer their NFT plus its stored value to another through a mutually agreed upon transaction. If a user sells or trades an NFT with another user, they are also transferring the full collateral amount of the NFT to the user they are transacting with.

14.1 Leaderboard

Users can compare their Zoo statistics with friends, family, and with every NFT holder globally. The leaderboard also allows users to make an offer bid on any NFT regardless of whether it is listed on the marketplace or not. Items are shown based on the NFT's total value or the user's cumulative NFT collection value.

15 Gen 0 NFT Drop

In the Gen 0 NFT drop, an enhanced reward system creates an engaging and rewarding experience for the Zoo community. Each Gen 0 animal NFT comes with its own unique set of daily rewards, designed to incentivize active participation and offer ongoing benefits to collectors. Whether additional \$ZOO tokens, exclusive NFTs, or other enticing in-game rewards, the daily rewards system ensures that there is always something to look forward to.

Boosts have a direct impact on the daily rewards. By adding collateral to the animals they own (known as “Feeding”), players can boost their daily rewards. These boosts amplify the potential rewards earned, providing an additional incentive for collectors to enhance the value of their animals over time.

The initial rewards are carved out during the Gen 0 NFT drop, with 50% of the entire token supply allocated towards rewards. This allocation sets the foundation for a robust and sustainable reward system. Any further rewards, reward modifications, or the duration of the reward period are subject to the decisions made by the Zoo DAO.

The daily rewards for each Gen 0 animal have a finite duration: daily rewards continue for a period of two years after the animal is minted. This time frame allows for a meaningful and engaging reward experience while maintaining a healthy ecosystem.

15.1 Breeding and Hatching Guide

Breeding Basics. Breeding requires two adult NFT animals of the same species but can be of any generation. The cost of breeding is determined by the original egg price (OEP), which varies based on rarity:

Egg Type	OEP
Sublime	1 ETH
Rare	0.5 ETH
Endangered	0.05 ETH

Upon successful breeding, an egg is minted and can be hatched for free or sold on the marketplace.

Growth Stages.

- Baby → Teen: lock in collateral equal to $1 \times$ OEP
- Teen → Adult: lock in collateral equal to $2 \times$ OEP
- Breeding fee: $3 \times$ OEP (directed to the DAO, not staked as collateral)

Generational Limits. Each egg generation has a limit of one baby, one teen, and one adult that can be minted. Adults with breeding privileges can breed with another adult of the same species to mint a subsequent egg generation. The first generation can breed 7 times, the second 6, and so on, until the seventh generation which has no breeding privileges.

15.2 Daily Reward Schedule

The rarity category of the animal determines its daily allowance based on age:

Rarity	Baby	Teen	Adult
Sublime	1 \$ZOO	2 \$ZOO	3 \$ZOO
Rare	0.25 \$ZOO	0.5 \$ZOO	1 \$ZOO
Endangered	0.05 \$ZOO	0.1 \$ZOO	0.2 \$ZOO

Breeding privileges are based on the generation of the animal, while the rarity category determines the daily allowance based on the age of the animal.

16 Asset Transfer

16.1 Unlocking Value via Asset Transfer Mechanisms

The Zoo Ecosystem offers inventive channels to realize the value of digital companions. The principal methods of asset transfer are trading NFTs on the marketplace and “burning” the NFT.

16.2 Trading on the Marketplace

A compelling means of unlocking a digital animal’s value lies in its sale on the marketplace. This platform provides a stage where the inherent value of an NFT, enriched by careful nurturing and potential rarity, can be transformed into tangible returns. The seller establishes a fair selling price, accounting for the accumulated value the NFT has amassed over time. Once the transaction is complete, the buyer inherits the animal’s NFT and the intrinsic value it has gathered, which encompasses the collateral fed to the animal and its earned rewards.

The exchange is more than a mere shift of ownership; it signifies the transference of a prospering legacy. The new owner steps into a realm where they can further cultivate the NFT and harvest the rewards from their new digital ally.

16.3 Burning NFTs

The Zoo Ecosystem introduces the concept of “burning” an NFT as an alternate route to tap into its value. This process mirrors the real-world practice of removing a coin or note from circulation. Burning marks the end of an NFT’s digital life cycle in the Zoo Ecosystem. This cessation of the NFT’s lineage and its potential yield is balanced by the opportunity to promptly access the NFT’s underlying earned value.

Burning an NFT provides an immediate way to extract the value nurtured within a digital companion, from feeding to daily rewards. While the action marks the finale of the pet’s digital journey, it culminates in a rewarding climax that underscores its lifecycle.

These mechanisms of trading or burning NFTs are instrumental in fostering a dynamic and vibrant Zoo Ecosystem. They ensure high liquidity, contribute to maintaining high value for NFTs based on their underlying assets, and enhance their inherent worth.

17 Zoo Animal Rewards

This section details the specific properties, rewards, and descriptions for each animal species in the Gen 0 drop.

17.1 Red Wolf (*Canis rufus*)

- **Generations Possible:** 1–7
- **Status:** Sublime
- **Daily Reward:** Baby: 1 \$ZOO, Teen: 2 \$ZOO, Adult: 3 \$ZOO
- **Original Egg Price:** 1 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 1 ETH (1× OEP); Mint Adult: 2 ETH (2× OEP); Breed Adult: 3 ETH (3× OEP)
- **Boosts:** 1.1–1.4×

Red Wolves are the rarest animals in the Zoo ecosystem, with only an estimated 35 or fewer individuals remaining in the wild. Derived from the Sublime Egg, the Red Wolf is highly coveted and earns the most daily rewards of any animal. The Red Wolf will only have 7 generations, with the first generation able to breed 7 times, and decreasing by one for each subsequent generation.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.2 Amur Leopard (*Panthera pardus orientalis*)

- **Generations Possible:** 1–7
- **Status:** Rare
- **Daily Reward:** Baby: 0.25 \$ZOO, Teen: 0.5 \$ZOO, Adult: 1 \$ZOO
- **Original Egg Price:** 0.5 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 0.5 ETH; Mint Adult: 1 ETH; Breed Adult: 1.5 ETH
- **Boosts:** 1.1–1.4×

The Amur Leopard is one of the rarest cats in the world, with an estimated population of under 70 individuals left in the wild. One of two animals derived from the Rare Egg, it is the second most rare in the ecosystem.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.3 Nubian Giraffe (*Giraffa camelopardalis*)

- **Generations Possible:** 1–7
- **Status:** Rare
- **Daily Reward:** Baby: 0.25 \$ZOO, Teen: 0.5 \$ZOO, Adult: 1 \$ZOO
- **Original Egg Price:** 0.5 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 0.5 ETH; Mint Adult: 1 ETH; Breed Adult: 1.5 ETH
- **Boosts:** 1.1–1.4×

The Nubian Giraffe is in danger with less than 2,645 individuals left in the wild lands of Ethiopia, Kenya, Uganda, South Sudan, and Sudan. Their population has dropped 40% in the last three decades due to habitat loss, conflicts with humans, civil unrest, and poaching. One of two creatures from the Rare Egg.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.4 Sumatran Elephant

- **Generations Possible:** 1–7
- **Status:** Endangered
- **Daily Reward:** Baby: 0.05 \$ZOO, Teen: 0.1 \$ZOO, Adult: 0.2 \$ZOO
- **Original Egg Price:** 0.05 ETH

The Sumatran Elephant is an endangered species native to the rainforests of Sumatra. One of four animals derived from the Endangered Egg.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.5 Javan Rhinoceros (*Rhinoceros sondaicus*)

- **Generations Possible:** 1–7
- **Status:** Endangered
- **Daily Reward:** Baby: 0.05 \$ZOO, Teen: 0.1 \$ZOO, Adult: 0.2 \$ZOO
- **Original Egg Price:** 0.05 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 0.05 ETH; Mint Adult: 0.1 ETH; Breed Adult: 0.15 ETH
- **Boosts:** 1.1–1.4×

The Javan Rhinoceros was once widely spread throughout Southeast Asia, India, and China, but is now critically endangered with only one known population in the wild and none in captivity. One of four animals derived from the Endangered Egg.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.6 Pygmy Hippo (*Choeropsis liberiensis*)

- **Generations Possible:** 1–7
- **Status:** Endangered
- **Daily Reward:** Baby: 0.05 \$ZOO, Teen: 0.1 \$ZOO, Adult: 0.2 \$ZOO
- **Original Egg Price:** 0.05 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 0.05 ETH; Mint Adult: 0.1 ETH; Breed Adult: 0.15 ETH
- **Boosts:** 1.1–1.4×

The Pygmy Hippopotamus is a rare and elusive creature found only in the dense forests of West Africa, mainly in Liberia, with small numbers in Sierra Leone, Guinea, and the Ivory Coast. One of four animals derived from the Endangered Egg.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.7 Siberian Tiger (*Panthera tigris altaica*)

- **Generations Possible:** 1–7
- **Status:** Endangered
- **Daily Reward:** Baby: 0.05 \$ZOO, Teen: 0.1 \$ZOO, Adult: 0.2 \$ZOO
- **Original Egg Price:** 0.05 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 0.05 ETH; Mint Adult: 0.1 ETH; Breed Adult: 0.15 ETH

- **Boosts:** 1.1–1.4×

The Siberian Tiger, also known as the Amur Tiger, is a magnificent creature native to the eastern regions of Russia. There are only around 500 left in the wild due to poaching, habitat loss, and hunting. One of four animals derived from the Endangered Egg.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

17.8 Emperor Penguin (*Aptenodytes forsteri*)

- **Generations Possible:** 1–7
- **Status:** Endangered
- **Daily Reward:** Baby: 0.001 \$ZOO, Teen: 0.01 \$ZOO, Adult: 0.1 \$ZOO
- **Original Egg Price:** 0.05 ETH
- **Mint Costs:** Hatch to Baby: Free; Mint Teen: 0.05 ETH; Mint Adult: 0.1 ETH; Breed Adult: 0.15 ETH
- **Boosts:** 1.1–1.4×

The Emperor Penguin is one of the most distinctive and largest penguin species. Their only habitat is Antarctica. They are threatened by climate change, which causes a significant loss of their icy home. One of four animals derived from the Endangered Egg.

Utility: Offers, Buy/Sell, Collateralize, Boosts, Earns, Create Pool, Breeding, Hatch Eggs, Burns, Metaverse.

18 Roadmap

At Zoo Labs, we are set on a journey towards creating a more vibrant, immersive, and interactive NFT universe. The roadmap for future upgrades includes features that will redefine the user experience and extend the scope of engagement with NFT animals.

18.1 Planned Features

- **Emotionally Intelligent Avatars:** NFT animals will evolve beyond static digital assets into dynamic, emotionally intelligent beings that users can interact with in the metaverse. With planned integration of AI, these animals will understand and reciprocate emotions, creating rich emotional connections.
- **Mini-Games and Metaverse Lands:** Enhanced gaming experience through introduction of mini-games and expanded metaverse lands. Special powers and play-to-earn games offer exciting new ways to engage and reap rewards.
- **Gold and Silver Coin Collection:** Through partnership with the Lux blockchain, players will have the opportunity to collect and redeem gold and silver coins, adding another layer of engagement and incentive.
- **New Animal Species:** Introduction of new animals, including extinct species and mythical creatures, further broadening the scope of the NFT animal universe.
- **Celebrity Collaborations:** Collaborations with celebrities and influencers for animal collections, creating cross-over between entertainment and digital collectibles.
- **Unique Animal Traits:** Introduction of unique animal traits such as horns, double heads, and wings, alongside a variety of skins and powers. Each NFT animal is as unique and individual as its owner.

- **Interactive VR/AR Pet App:** An interactive VR/AR pet app that responds emotionally and serves as a personalized AI-powered emotional support companion.
- **DAO Governance:** Formation of a DAO that will oversee all game functions, manage the treasury, forge partnerships, and allocate rewards for each animal.

18.2 Future Game Interface

The Zoo Labs gaming landscape consists of floating islands in the vast cosmos, each distinct and teeming with life. Each island mirrors the unique attributes of the NFT animals inhabiting them, creating diverse ecosystems.

Users are given the opportunity to embody their NFT animals, experiencing the world through their eyes. They engage in play-to-earn games, navigate through captivating environments, and participate in quests and treasure hunts. The Zoo Labs universe is a thriving social space where users can connect with friends and other players from around the globe.

Intelligent avatar animals serve as AI-powered companions that bring a sense of social engagement. These avatars, capable of learning, adapting, and reacting, interact with users organically.

18.3 Play-and-Earn Games

Coin Runner. An endless-runner style quest for valuable gold and silver coins through the Zoo Labs temple. Players choose one of their Zoo Animals to play with, each with unique abilities and skills. They dash through winding paths, collecting gold and silver nuggets of various denominations, from small nuggets to the coveted bullion bar. Collected coins can be used to unlock power-ups, upgrades, and new animals. All rewards are redeemable for physical assets or can be transferred on Zoo Market or a secondary marketplace.

Memory Bridge. A memory-based game set in the Zoo Labs Metaverse. Players guide their Zoo Animal across a treacherous bridge by memorizing the pattern of bricks. The bridge becomes longer as the game progresses, increasing the challenge and potential rewards. Players have 4 seconds per level to step on the correct brick. Successfully completing each level earns valuable rewards, including rare items, tokens, and special powers. Every reward collected is redeemable for real-world assets.

19 Zoo DAO

At the heart of the Zoo Protocol lies the Zoo DAO, a community-driven initiative dedicated to preserving endangered species from the brink of extinction. The DAO offers rewarding experiences and empowers members to shape the future of the Zoo Protocol through meaningful decisions.

Zoo DAO governs the Zoo Protocol, a sophisticated set of smart contracts for collateral-backed NFT Animal assets. When users join the Zoo DAO, they earn Experience Points (XP) and governance tokens (KEEPER), empowering them to participate actively in the protocol.

19.1 Experience Points and Leveling

The accrual of XP tokens is tied to undertaking “quests” or tasks, each with a specific reward allocation of XP and KEEPER tokens. These quests vary in complexity, with higher-level tasks requiring more substantial contributions to the community and yielding more XP. The more XP

earned, the faster one advances through the ranks. Level is determined by the equation:

$$\text{Level} = \text{level} \times 300 - 500 \quad (4)$$

Progressing through levels grants greater authority within the ecosystem and broadens influence. XP tokens are non-transferable and strictly tied to Decentralized Identification Services (DIS) and comprehensive KYC procedures, preserving the integrity of the system.

19.2 Governance and Voting

Votes are open for all members; however, some voting privileges are reserved for members of certain levels. Generally, higher-level votes are more privileged and controlling decisions. Specific actions such as proposing, voting, and interacting with governance mechanisms may require a blend of \$ZOO and KEEPER tokens. Different proposals cater to different levels, underlining the ecosystem’s commitment to inclusivity.

19.3 Betting-Based Governance

Central to the Zoo Governance and Voting Protocol is the vZOO level and the stake age duration. Unique to the system, KEEPER token holders “bet” on proposals rather than voting. Winning bets yield additional tokens, promoting active participation. The process allows anyone who meets the requirements to submit a proposal, with those gaining traction advancing to the next round, while lesser-supported ones are eliminated. This distinctive approach ensures a vibrant and thriving ecosystem, rewarding active participation and fostering continuous community involvement.

20 Zoo: An NFT Liquidity Protocol

Zoo is a liquidity protocol for NFTs, in the way that Uniswap or PancakeSwap is for tokens.

New animals are added to the Zoo ecosystem by deploying new “Drop Contracts,” which define the Animal NFTs content. Once an Egg NFT hatches, an Animal NFT of the same status is minted and begins to earn \$ZOO Tokens according to its staked collateral and the outlined daily reward, determined by its rarity and species.

Based on rarity, age, and how much collateral is added to each NFT, each NFT is effectively a liquidity position (LP) that can be burned to receive the underlying assets (collateral + rewards earned).

Before freeing an animal (burning the NFT), \$ZOO and practically any currency can be added as liquidity to the animal NFT to increase its value. When users add collateral to their NFT Animal or participate in transactions on the Zoo ecosystem, a small fee is taken to be donated to the Zoo Labs Foundation. The economics have been designed to cater to the interests of collectors and charities alike while slowly supporting buy pressure through the gamification of the Zoo game, increasing the value of the total ecosystem.

20.1 Pools

Users can list NFTs as a pool on the marketplace. By placing NFTs in a pool, users can determine the value of the entire set of NFTs by setting the price of the pool. Once a pool is created, other users can purchase the pool for the set value. The buyer must send the amount to the owner to effectively transfer ownership.

21 NFTs That Make You Smile

Zoo Labs is forging a revolution in the NFT space, bringing not just awe-inspiring visuals, but also genuine emotional connections into the digital domain. The goal transcends the standard expectations of NFT collectibles, with the integration of cutting-edge AI, creating intelligent avatars nestled within virtual animal NFTs.

Enriched with emotional intelligence, these NFTs evolve beyond mere digital assets, emerging as lively companions capable of recognizing and responding to emotions. They offer engaging dialogues, personalize themselves to preferences, and deliver companionship reminiscent of an emotional support pet living in a digital ecosystem.

Built on advanced AI and intricate language models, the intelligent avatars grow and adapt over time, cultivating distinct personalities and understanding unique needs. They lend an attentive ear, shower words of encouragement, and offer assistance in daily tasks, truly becoming digital allies. Whether it is a casual chat or a comforting presence one seeks, the intelligent avatar NFTs are always there in digital escapades.

21.1 AI-Powered Language Model Integration

At Zoo Labs, cutting-edge AI language models are unified with virtual animal NFTs, redefining the traditional gaming experience with a deeper emotional resonance. Digital creatures become dynamic companions, engaging in meaningful conversations, memorizing interactions, and lending a hand, much like a supportive digital pet.

By leveraging the power of AI and its APIs, virtual animal NFTs morph into dynamic companions, understanding context and personalizing experiences. These virtual pets converse, adapt, and companion users whenever needed. They are empathetic, supportive, and ready to listen and assist in daily tasks.

This unique blend of AI and virtual creatures separates Zoo from conventional NFTs, facilitating a personalized and interactive experience that expands beyond the norms of traditional gaming. With continuous AI advancements and state-of-the-art language model integrations, the intelligence and abilities of avatar animals are continually enhanced, facilitating the formation of genuine relationships with virtual companions in a vibrant world where virtual animals spring to life.

22 Bridging Blockchains

As the DeFi space continues to experience exponential growth, the number of blockchains available for investment also increases. However, each blockchain has its characteristics, with some being easy to use but expensive in fees, while others have light fees but are complex to navigate. The Zoo Bridge Protocol addresses these limitations by developing a bridge between major blockchains, facilitated by \$ZOO. This bridge, established through the Zoo platform, facilitates universal exchange within the blockchain space.

22.1 Threshold Signature Scheme

While traditional bridges trap tokens in vault contracts, the Zoo Bridge Protocol implements a unique approach, leveraging a threshold signature scheme. Oracles monitor the blockchain network to ensure funds are deposited in the smart contract, with ZOO tokens being burned on one chain

via the bridge protocol and minted on the desired chain. This eliminates the need for a vault contract that could potentially contain vulnerabilities that could be exploited by attackers.

The Zoo Bridge is entirely secured by cross-chain multi-party computation (MPC) validators who operate with threshold signature schemes (TSS).

22.2 Leaderless Consensus

The network is leaderless and ensures security through each validator running the same process upon receiving on-chain events on the various chains that the MPC validator group tracks. Consensus is achieved when two-thirds of all validators collectively sign the same transaction using their key, and a trade is issued to the destination chain. The \$ZOO is minted when enough oracles see the deposit into the smart contract. In a standard transaction, such as when ZOO teleports from one chain to another, three network confirmations are awaited.

The Zoo Bridge Protocol represents a significant innovation in blockchain technology, improving the interoperability of blockchains and expanding investment opportunities.

23 Open Source

By making Zoo open-source, developers can build on top of Zoo and use the \$ZOO token as their projects' virtual utility token or build other games and NFT platforms parallel to the codebase.

This allows for the utility of the \$ZOO token to go beyond that which the project can support itself and become the virtual utility token of a potentially endless number of additional projects, creating enormous potential that is not possible without alignment to open-source ideology. Furthermore, the project's strong foundational support in the field of Education is another element that will make it shine through the use of open-source elements. This content will all be created and curated by the Zoo Foundation's resident Director of Education.

Zoo will be working with partners in conservation to establish content to be shared through the Zoo Foundation in an open-source format. This educational content, aligned to nationally recognized learning standards for grade levels K-12 and supplementary content, will be completely free to schools around the US. Schools will be encouraged to share the results of their time spent working with the materials in a repository of open-source materials found on the Zoo Foundation's website.

The development team is dedicated to improving the future of the world. The possibilities are endless and begin with community engagement.

24 Partnerships

At Zoo Labs, we firmly believe in the power of collaboration and community-driven initiatives. As we pave the way for a brighter future of conservation and environmental stewardship, we are excited to share our plans for future partnerships, upgrades, and game features that will fuel the growth of the Zoo Labs community.

24.1 Collaborating with Non-Profits

Our commitment to protecting endangered species and preserving their habitats extends beyond the boundaries of Zoo Labs. We are actively seeking partnerships with esteemed non-profit organizations that share our mission and values. Through these collaborations, we aim to leverage

collective strengths to drive impactful change. By combining resources, expertise, and networks, we can amplify efforts in fundraising campaigns, educational programs, and joint conservation projects.

24.2 Empowering the DAO

The decentralized autonomous organization (DAO) lies at the core of Zoo Labs, empowering community members to actively participate in shaping the future of the platform. Through the DAO, community members have a voice in decision-making processes, including selecting future partnerships, voting on game upgrades, and influencing the direction of Zoo Labs. Decentralized governance fosters inclusivity, transparency, and collective ownership. By placing the power in the hands of the community, we unlock the potential for self-sustainability and create an ecosystem where everyone's contributions matter.

24.3 Continuous Game Enhancements

The dedicated team of developers and designers are continuously working to enhance the Zoo Labs game experience. We are committed to introducing exciting upgrades that will captivate and engage players. Expect the introduction of new animal species, expanded habitats, and interactive gameplay elements that will immerse users in a world of adventure and discovery.

24.4 Cross-Platform Integration

As we strive to build a thriving community, we recognize the importance of expanding our reach and connecting with diverse ecosystems. We are actively exploring opportunities for cross-platform integration, collaborating with other blockchain-based gaming platforms, virtual reality experiences, and even real-world events. By forging these connections, we extend our community and provide players with a truly interconnected experience that transcends boundaries.

24.5 The Power of Decentralization

Zoo Labs embraces the decentralized nature of blockchain technology and recognizes its transformative potential. By leveraging decentralization, we foster inclusivity, transparency, and shared ownership within our community. The decentralized model ensures that decisions are made collectively, without the need for intermediaries or centralized control. This empowers individuals to contribute, govern, and shape the future of Zoo Labs.